THE POINT OF A LOCALIZED SITE INDUCED BY A POINT AND AN ELEMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Points of a site as morphisms from the topos of a point. Same chapter, Lemma lemma-site-point-morphism. Represents the given point by a set-valued functor and identifies the topos of a point with the topos of the associated category of sets.
- Localizing a morphism of sites. Same chapter, Lemma lemma-localize-morphism. Produces the localized morphism of sites fitting into the commutative diagram of topoi.

Source reference. Original lemma 04EL: sources/stacks-sites/pinned-original-source.tex, source lines 8908–8908. Navigation pointer invoked at proof line 8908; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03CF: sources/stacks-sites/pinned-original-source.tex, source lines 8914–8914. Navigation pointer invoked at proof line 8914; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XO: sources/stacks-sites/pinned-original-source.tex, source lines 8953–8953. Navigation pointer invoked at proof line 8953; the counted

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Source reference. Original section 00XW: sources/stacks-sites/pinned-original-source.tex, source lines 8955–8955. Navigation pointer invoked at proof line 8955; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let p be a point of $\mathcal{C}$ given by $u : \mathcal{C} \to Sets$. Let U be an object of $\mathcal{C}$ and let $x \in u(U)$. The functor*

$$v : \mathcal{C}/U \longrightarrow Sets, \quad (\varphi : V \to U) \longmapsto \{y \in u(V) \mid u(\varphi)(y) = x\}$$

defines a point q of the site $\mathcal{C}/U$ such that the diagram

$$\begin{array}{ccc} & & Sh(pt) \\ & \overset{q}{\swarrow} & \downarrow p \\ Sh(\mathcal{C}/U) & \xrightarrow{j_U} & Sh(\mathcal{C}) \end{array}$$

commutes. In other words $\mathcal{F}_p = (j_U^{-1}\mathcal{F})_q$ for any sheaf on $\mathcal{C}$.

Proof. Choose S and $\mathcal{S}$ as in Lemma 04EL. We may identify $Sh(pt) = Sh(\mathcal{S})$ as in that lemma, and we may write $p = f : Sh(\mathcal{S}) \to Sh(\mathcal{C})$ for the morphism of topoi induced by u. By Lemma 03CF we get a commutative diagram of topoi

$$\begin{array}{ccc} Sh(\mathcal{S}/u(U)) & \xrightarrow[j_{u(U)}]{} & Sh(\mathcal{S}) \\ \downarrow{\scriptstyle p'} & & \downarrow{\scriptstyle p} \\ Sh(\mathcal{C}/U) & \xrightarrow{j_U} & Sh(\mathcal{C}), \end{array}$$

where p' is given by the functor $u' : \mathcal{C}/U \to \mathcal{S}/u(U)$, $V/U \mapsto u(V)/u(U)$. Consider the functor $j_x : \mathcal{S} \cong \mathcal{S}/x$ obtained by assigning to a set E the set E endowed with the constant map $E \to u(U)$ with value x. Then j_x is a fully faithful cocontinuous functor which has a continuous right adjoint $v_x : (\psi : E \to u(U)) \mapsto \psi^{-1}(\{x\})$. Note that $j_{u(U)} \circ j_x = \mathrm{id}_{\mathcal{S}}$, and $v_x \circ u' = v$. These observations imply that we have the following commutative diagram of topoi

$$\begin{array}{ccccc} Sh(\mathcal{S}) & & \xrightarrow{\quad} & & \\ & \overset{a}{\searrow} & & & \\ & & Sh(\mathcal{S}/u(U)) & \xrightarrow[j_{u(U)}]{} & Sh(\mathcal{S}) \\ & \overset{q}{\swarrow} & \downarrow{\scriptstyle p'} & & \downarrow{\scriptstyle p} \\ & & Sh(\mathcal{C}/U) & \xrightarrow{j_U} & Sh(\mathcal{C}) \end{array} \qquad p$$

Namely:

(1) The morphism $a : Sh(\mathcal{S}) \to Sh(\mathcal{S}/u(U))$ is the morphism of topoi associated to the cocontinuous functor j_x, which equals the morphism associated to the continuous functor v_x, see Lemma 00XO and Section 00XW.
(2) The composition $p \circ j_{u(U)} \circ a = p$ since $j_{u(U)} \circ j_x = \mathrm{id}_{\mathcal{S}}$.
(3) The composition $p' \circ a$ gives a morphism of topoi. Moreover, it is the morphism of topoi associated to the continuous functor $v_x \circ u' = v$. Hence v does indeed define a point q of $\mathcal{C}/U$ which fits into the diagram above by construction.

This ends the proof of the lemma. □